Climate policy analysis with endogenous preference change and innovation

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Outline

1. Introduction
2. The model
3. Calibration and validation
4. Simulation results

Motivation

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  - Aggregate outcomes caused by technological paradigms can modify preferences.
- Overlooking the synergies between both outcomes may lead to sub-optimal policy-mixes.

Previous contributions

- **Case studies**
  - Windrum (2005) finds that under heterogeneous preferences multiple stages of product innovation may occur, culminating in permanent design heterogeneity.
  - Tripsas (2008) and Windrum (2009) finds that sudden changes in consumer preferences can cause new technological paradigms.
- **Simulation models**
  - (Adner, 2001; Windrum, 2009; Hermann, 2017) address how consumer preference heterogeneity alters the direction of innovations.
  - Stummer (2015) addresses how stochastic innovation shape the preferences of consumers in the Austrian Biofuel market.
  - Roos (2018) and Moore (2022) proposes a system dynamics "earth-world" model where aggregate outcomes influence opinion dynamics.
  - Konc et al. (2021) finds a "social multiplier" that decreases the optimal carbon tax.

This project

Research questions

- What is the interaction effect between innovation and opinion dynamics in total emissions?
- How does the interaction term affect the optimal carbon tax?

Method

- Partial dis-equilibrium agent-based model with endogenous consumption preferences (OD model) and endogenous innovation (NK model).
- Case study applied to the automobile industry ($\sim$20%) of total emissions.

The scale of analysis

Aggregate variables of interest:

- Industry total emissions.
- Weighted average vehicle fuel efficiency ("green productivity").
- Weighted average vehicle cost ("normal productivity").
- Market concentration.

The scale of analysis is meso-economic (between macro and micro).

- Marrero et al. 2018: Macroeconomic example, DSGE (with heterogeneous agents) for EV adoption.
  - Includes interactions with the macroeconomic environment.
  - Aggregates agents and omits their interactions.
- Naqvi 2024: Micro-simulation example, Simulation of individual mobility decisions in Vienna for emissions exposure.
  - Disentangles mobility decisions to a comprehensive "menu" of commutes.
  - Takes supplied car variants as given.

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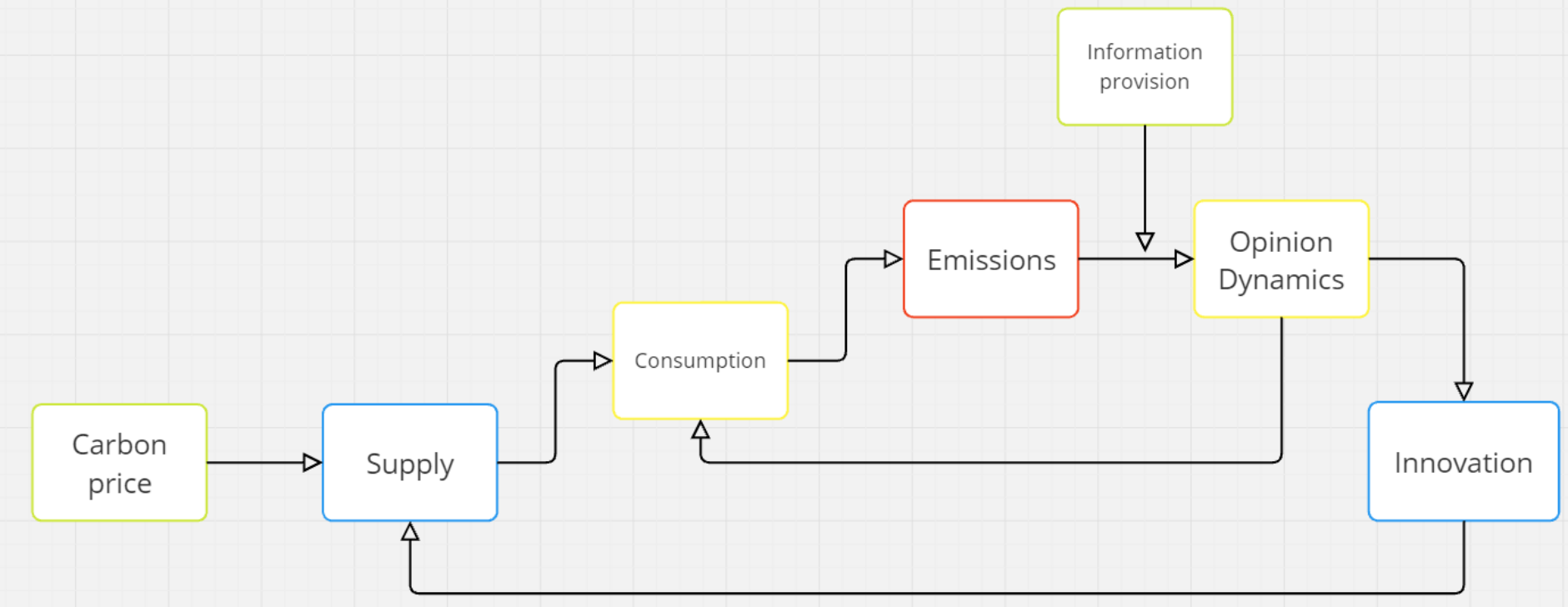
Model structure



Channels on the watch

Individual channels

- Innovation can reduce the overall emissions intensity of supplied goods, or increase the relative competitiveness of low-carbon goods.
- Opinion dynamics can increase the preferences for low-carbon goods and hence their consumption.

Interaction effects

- Changes in consumer preferences motivate changes in the direction of innovation, from cost reductions to emissions reductions.
- Innovation-caused increases in total emissions can increase climate awareness.

Consumption

1. Individuals observe the vehicles supplied Γ_t and evaluate their characteristics $\vec{X}_{g,t}$.
2. They consider buying a vehicle $g \in \Gamma_t$ with probability

$$\Phi(g' \in \Gamma_t, \Gamma_t, \vec{\beta}_{i,t}) = \frac{U^B_{i,t}(g')^{\kappa}}{\sum_{g=1}^{\Gamma_t} U^B_{i,t}(g)^{\kappa}} \tag{1}$$

   Where utility is a linear combination of price, quality and an environment score.
3. Buy vehicle if the utility is higher than the depreciated utility of the current car (facing no cost).

Opinion dynamics

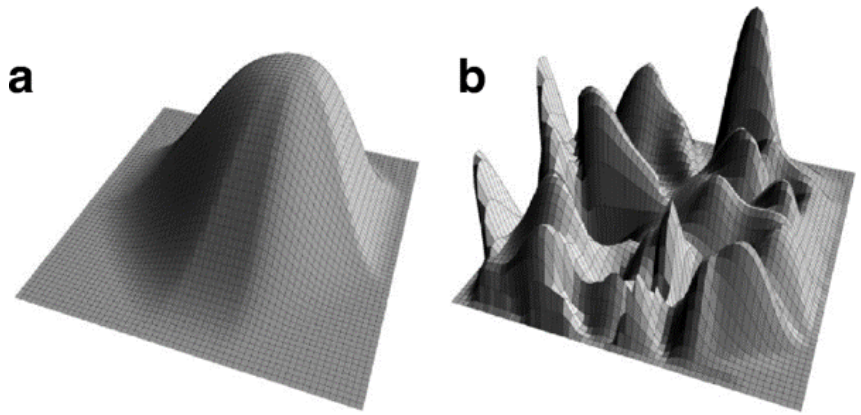
Individuals' opinion evolve with social influence and the perception of climate danger

$$\beta_{2,i,t} = (1-\upsilon)\beta_{2,i,t-1} + \upsilon\left((1-\upsilon_{E,i})S_{i,t} + \upsilon_{E,i}\frac{E_t}{E_{max}}\right) + \epsilon_t, \quad (2)$$

1. Current opinion
2. Social imitation
3. Emissions impact

Technologies: Fitness landscape

Firms locally search for technologies over an *NK* landscape (Kaufman, 1993).
Each landscape location return the product attributes (cost, quality, environment score).



Low K surface High K surface

Innovation: Search process

We use Jain & Kagut (2014) memory with neutrality algorithm.

1. Firms learn the population in market segments
2. Evaluate the expected profit of all neighbouring alternative technologies.
3. Classify technologies based on expected profit.
4. Acquire one technology of the same or higher rank

Evaluation function:

$$\Pi(v \in V, \Gamma \subset V, |\vec{I}_{w,t}|) = \mu X_v^C \sum_{w=1}^{W} |I_w| \Phi(v \in V, \Gamma \subset V, \vec{\beta}_w) \quad (3)$$

- Per unit profit.
- Population in market segment
- Expected in-segment market share

Supply

1. For each market segment $w : |I_w| \geq Y$, each firm $j \in J$ offers a car design.
2. Price equals costs plus an exogenous fixed mark-up.
3. **Carbon price**: Add $\tau(1 - X^C_{g,t})$ to the price.

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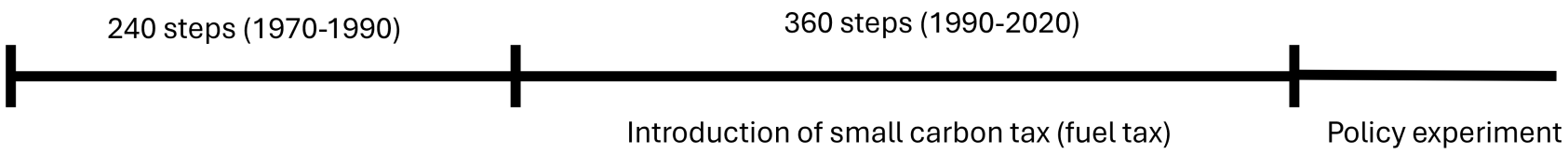
Calibration strategy

1. Parameters obtained from scientific articles: exogenous preferences (Ostil et al., 2017), industry mark-up (Grieco et al., 2024), number of firms (Benkard et al., 2023), or carbon price (Aghion, 2016).
2. The non-observable parameters are calibrated indirectly to replicate documented trends of the automobile industry.
   - **T1**: Rise-and-fall trend in market concentration (Grieco et al., 2024).
   - **T2**: Reduction in costs and increasing quality followed by rising fuel efficiency and unchanged cost and quality (Grieco et al., 2024).
   - **T3**: Persistent attribute heterogeneity among vehicles (Windrum, 2009).
   - **T4**: Increasing total emissions followed by a small decline in the last 4 years (OurWorldinData).



Validation: Market concentration

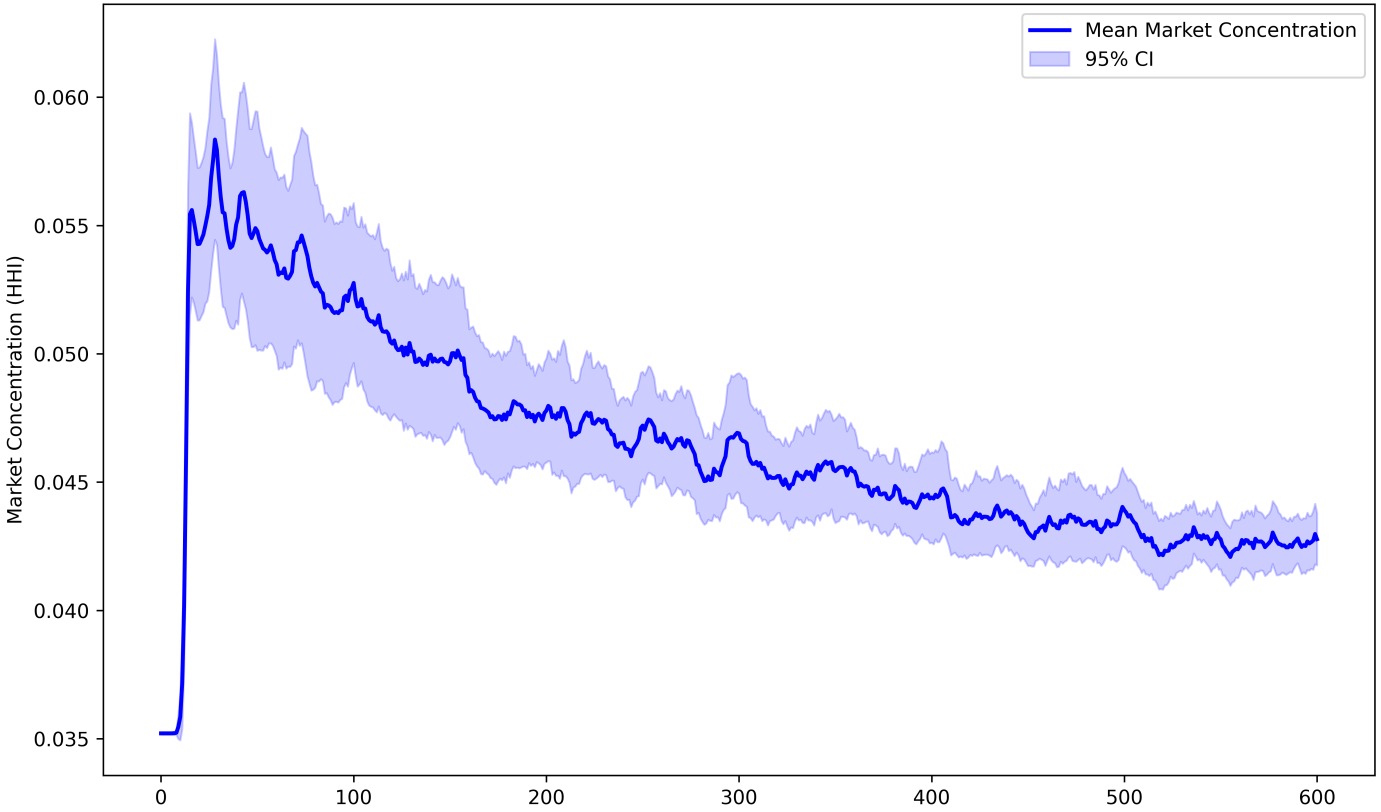


Figure: **T1**: Rise-and-fall trend in market concentration (Grieco et al., 2024)

Validation: Attributes

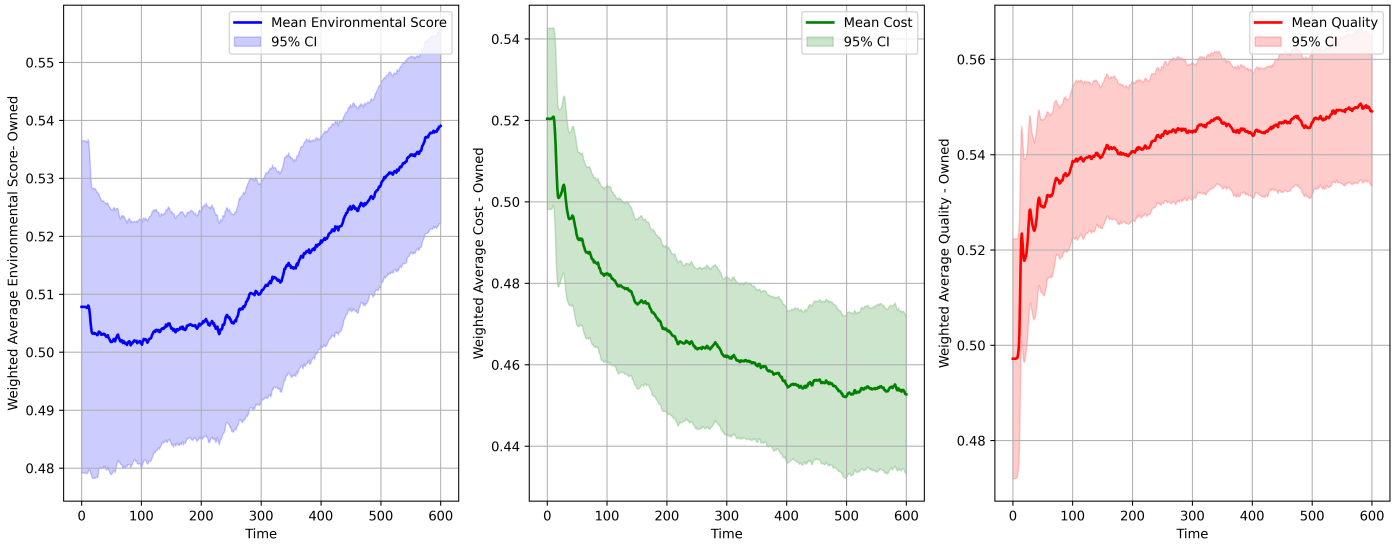


Figure: **T2**: Reduction in costs and increasing quality followed by rising fuel efficiency and unchanged cost and quality (Grieco et al., 2024)

Validation: Heterogeneity

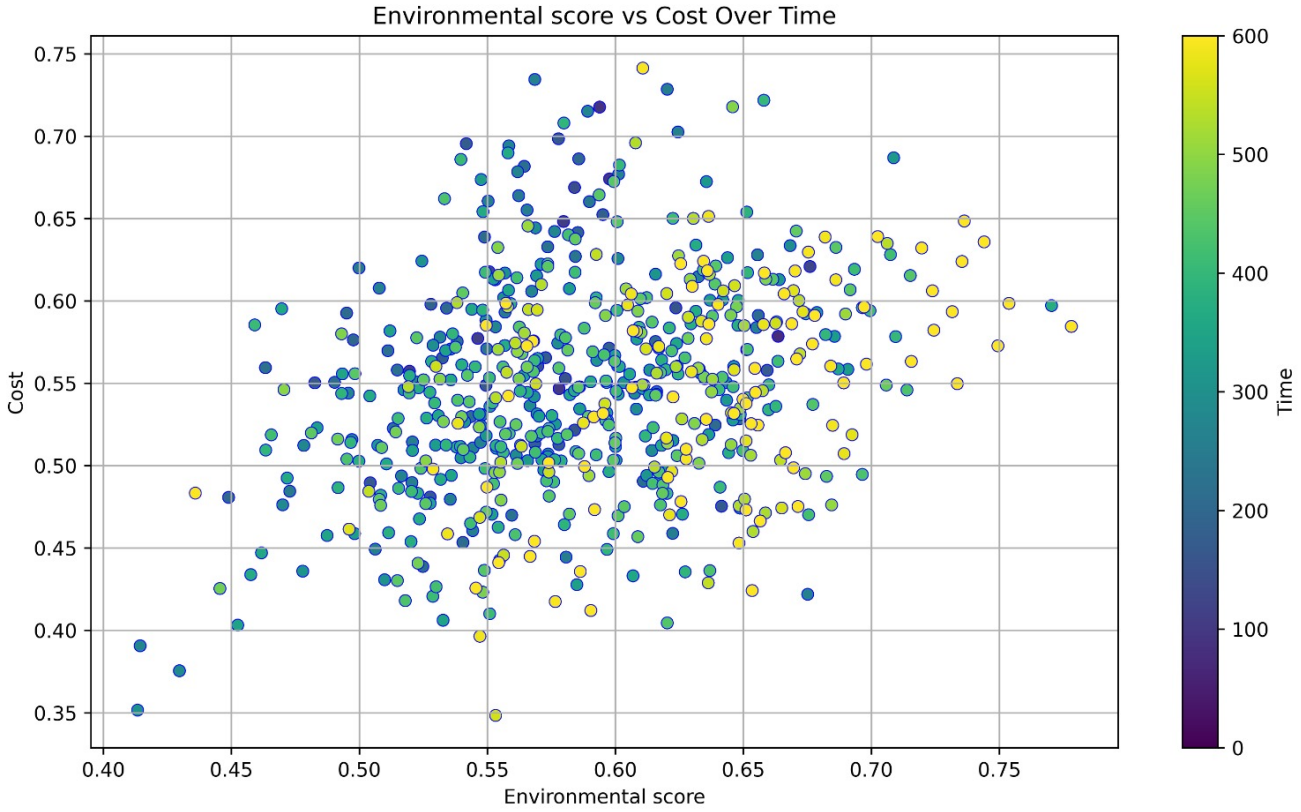


Figure: **T3**: Persistent attribute heterogeneity among vehicles (Windrum, 2009))

Validation: Emissions

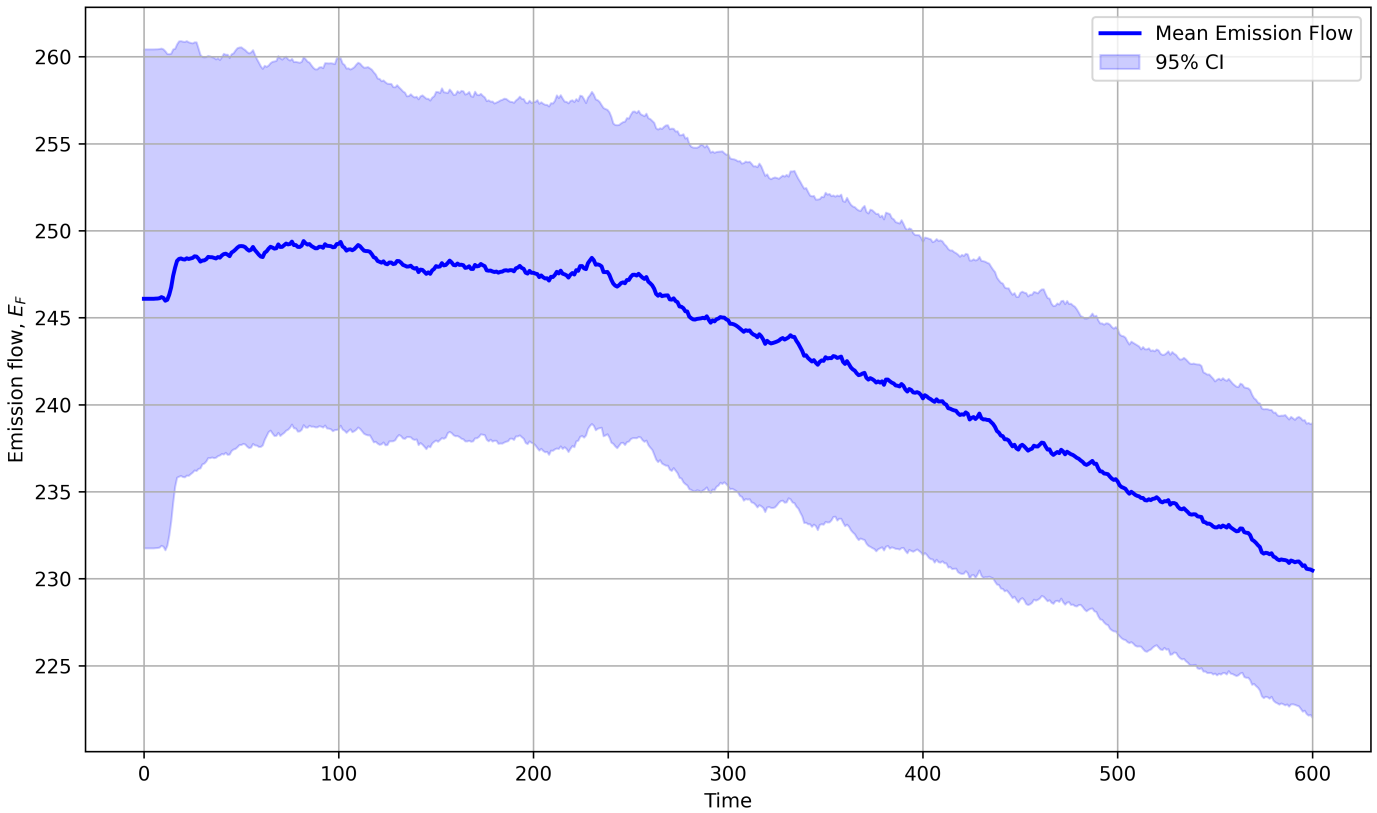


Figure: **T4**: Increasing total emissions followed by a small decline in the last 4 years (OurWorldinData).

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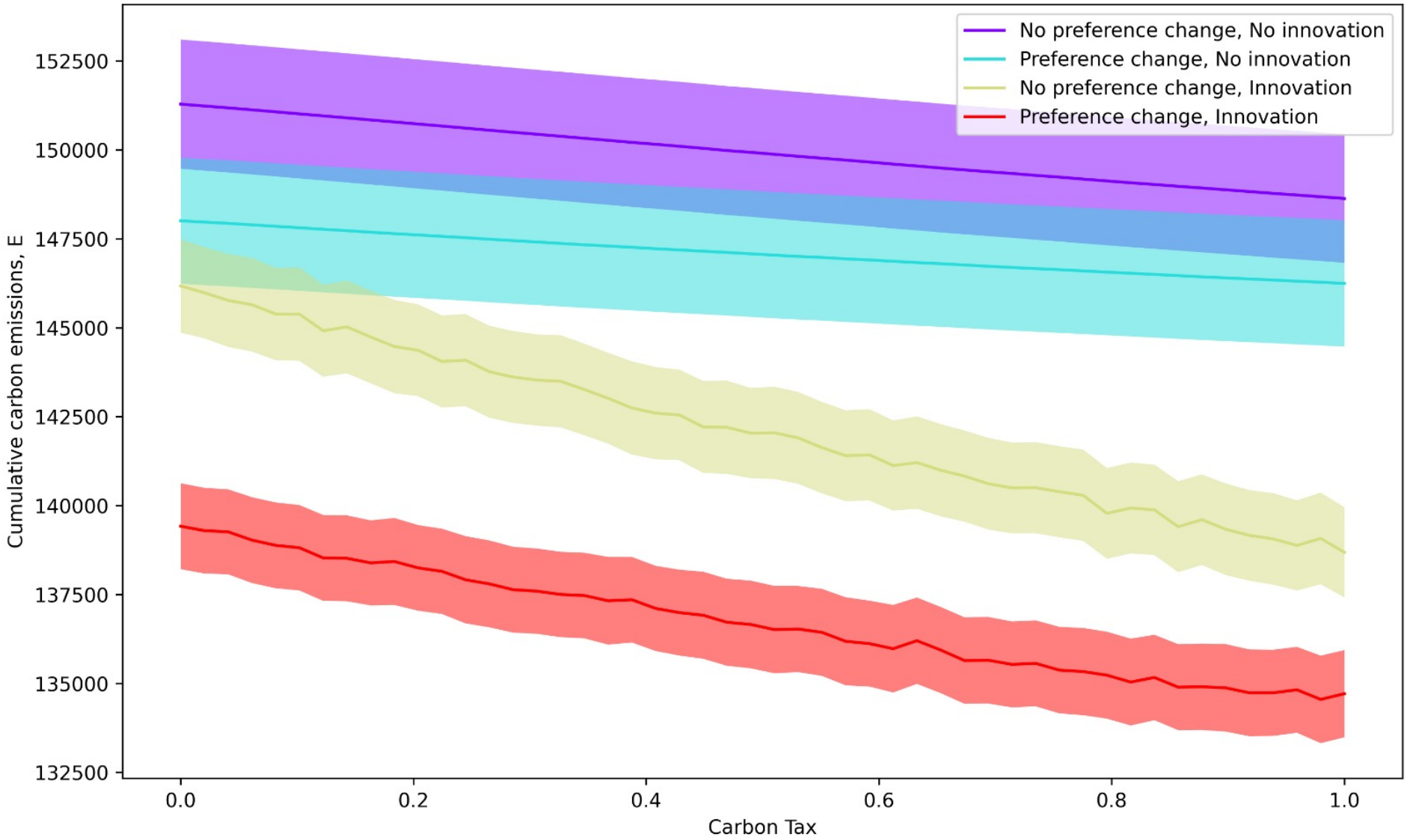
Approach

- RQ: What is the interaction effect between innovation and opinion dynamics?
- We simulate 400 runs for different carbon price values and four cases:
  1. No preference change and no innovation.
  2. Preference change and no innovation.
  3. No preference change and innovation.
  4. Preference change and innovation.
- Perform a simple OLS regression to find the marginal effects on last time-step cumulative emissions.

Information provision: The experiments so far were inconclusive.

Results


No preference change, No innovation
Preference change, No innovation
No preference change, Innovation
Preference change, Innovation
Cumulative carbon emissions, E
152500
150000
147500
145000
142500
140000
137500
135000
132500
0.0
0.2
0.4
0.6
0.8
1.0
Carbon Tax

Results and intuitions

- There is an interaction effect only in levels: Changes in preferences speed innovation re-direction.
- Lower tax effect on emissions when considering opinion dynamics: substitute effects (concerned individuals are price-insensitive).
- When considering both channels, the carbon tax associated with an emissions target is always lower.

Concluding remarks

- We study the interaction between preference change and innovation in the context of the green transition in the automobile industry.
- We propose a partial dis-equilibrium agent-based model where both phenomena interact.
- We find a significant emission reduction interaction effect.
- Endogenous preferences act as a substitute for carbon taxes.
- Considering both channels simultaneously leads to a lower prescribed carbon tax.

Cool... but cool enough?

(...)"the "gold rush" era of CAS modeling in strategy and organization science is over. The general CAS-based metaphors have been absorbed in the literature, the basic mechanisms underlying the major modeling structures have created their "aha" effects, and the buzz around "complexity" (broadly speaking) has faded." (...) "it might be time to get down into the organizational "engine room" and get our hands dirty, developing models of organizations as CAS that are somewhat more complex." (Baumann, 2015. pg. 181)

Currently working on...

- Calibrate to validate the model quantitatively.
- Testing alternative social learning models with imperfect information.
- Split into a 2-step decision: which car I buy? How much distance I travel? (Necessary to introduce public transportation and generate rebound effects).
- Distinguish different types of cars with unique *NK* landscapes (Diesel, gasoline, EVs).
- Experiment alternative policy experiments: information provision and public transportation.

Thank you!

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